COS 487 / 587 Course Project Part 2: IR Rangers

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Abstract

Keywords

information retrieval, BM25, SPLADE, dense retrieval, rank fusion, scientific document retrieval, passage retrieval

1 Introduction

The volume of scientific literature makes finding relevant work increasingly difficult, both at the level of whole papers and of the specific passages that answer a question. In the first phase of this project, we built a retrieval system for arXiv papers and paragraphs[1] using two term-matching models, and found that BM25 significantly outperformed TF-IDF on both tasks. Our analysis also exposed the limits of exact term matching: hard queries that paraphrase technical terminology in general language, for example asking about "wrist-worn optical sensors" rather than PPG. share few terms with the document that answer them, and BM25 often failed on them, especially for paragraphs.

This paper describes the second phase, in which we add two neural retrieval models intended to bridge this vocabulary gap: a learned sparse model, SPLADE [5], which represents text as weights over vocablulary terms, including related terms the text does not contain, and a dense bi-encoder [10], which represents text as a single embedding vector. We evaluate each model both as released and after fine-tuning on our Train queries, rank papers by their best-scoring paragraph [4], and combine all three models with reciprocal rank fusion [3]. BM25, our best model from part 1, serves as the baseline throughout.

The test collection consists of 34,308 arXiv papers published in August and September 2024. Each paper is provided both as raw HTML scraped from the ar5iv website and as a processed JSON file containing its identifier, title, abstract, and paragraphs. Because the JSON files were produced with rule-based extraction, they contain noise such as leftover LaTeX markup and merged words, and display equations are missing altogether. In total, the collection contains 2,275,574 paragraphs, an average of about 66 per paper.

We address two tasks over this collection: *paper retrieval*, which returns papers identified by their arXiv ID, and *paragraph retrieval*, which returns individual paragraphs. For each query, systems return their top 100 results. Three query sets are provided: Train (2,000 queries), Study (250 queries with relevance judgements, which we use for all analysis in this paper), and Test (250 queries whose judgments are withheld). Each set contains 30% easy, 50% medium, and 25% hard queries. Relevance is graded on a three-point scale: a document that fully addresses the query is relevant (2), one that provides useful information without fully addressing it is partially relevant (1), and any document not in the judgments is considered non-relevant.

[1] Code: https://github.com/esthergreene/IR-Rangers

SPLADE is the most effective single model on both tasks. On papers it improves nDCG@5 over BM25 from 0.514 to 0.631; on paragraphs its advantage is smaller and not significant. The dense model outperforms BM25 on papers but not on paragraphs, and fine-tuning did not significantly improve either neural model. Fusing all three models gives the best results, with nDCG of 0.701 on papers and 0.550 on paragraphs, significantly above both BM25 and SPLADE alone. Our analysis shows that the neural models can retrieve paragraphs that paraphrase a query without sharing its words, yet they still rank highly some paragraphs that only share its vocabulary.

2 Methodology

Our system retrieves with three first-stage models that share the query sets, run format, and evaluation pipeline. BM25 (Section 2.2) searches inverted indexes over separately preprocessed paper and paragraph collections, as in Part 1. The two neural models, SPLADE (Section 2.4) and a dense bi-encoder (Section 2.5), receive more lightly preprocessed text (Section 2.1) and index paragraphs only. Both start from publically released checkpoints, which we evaluate as released and after fine-tuning on Train queries (Section 2.3). Because the neural indexes contain only paragraphs, papers are ranked from their paragraphs' scores (Section 2.6). Finally, the runs from different models can be combined with reciprocal rank fusion (Section 2.7). Section 2.8 describes our evaluation setup.

2.1 Preprocessing

BM25 and the neural models receive differently preprocessed text, because they match text in different ways.

For BM25, we keep the Part 1 pipeline unchanged. Inline mathematics, remaining LaTeX commands, URLs, and numeric citation markers are removed; text is lowercased and split into alphanumeric tokens; tokens shorter than two characters or consisting only of digits are discarded; stopwords are removed using NLTK's English list [2], extended with terms common in scientific writing such as "et", "al", and "fig"; and the remaining tokens are reduced with the Snowball stemmer [9]. Cleaning reduces the paragraph collection from 168.0 to 114.9 million tokens (31.6%) and leaves 8,389 paragraphs, almost all standalone formulas, empty. These are excluded, leaving 2,267,185 indexed paragraphs. Both encoders trucate their input at 256 WordPiece tokens.

2.2 BM25 Baseline

Our baseline is the BM25 model from Part 1, unchanged. It scores a document d for a query q as

$$\text{score}(q,d) = \sum_{t \in q} idf_t \cdot \frac{tf_{t,d}\,(k_1+1)}{tf_{t,d} + k_1\left(1 - b + b\,L_d/L_{avg}\right)}, \qquad (1)$$

$$idf_t = \ln\left(1 + \frac{N - df_t + 0.5}{df_t + 0.5}\right), \tag{2}$$

where N is the number of documents, df_t the number containing term t, $tf_{t,d}$ the occurrences of t in d, and L_d and L_{avg} the length of d and the average length, in tokens. This is the "Lucene (accurate)" variant of Kamphuis et al. [7], whose IDF remains positive for terms that occur in more than half of the documents. The parameter k_1 controls how quickly repeated occurrences of a term saturate, and b how strongly scores are normalized by document length. Scoring is term-at-a-time over the postings of the query terms, vectorized with NumPy.

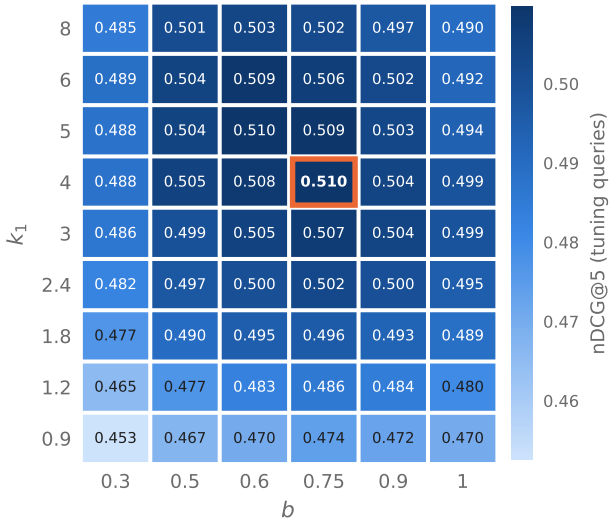
We tuned k_1 and b separately for each collection by grid search on 1,600 Train queries, selecting the setting with the highest nDCG@5 (Figure 1). The selected settings, $k_1 = 4.0$, $b = 0.75$ for papers and $k_1 = 0.4$, $b = 0.9$ for paragraphs, also outperformed both the textbook values ($k_1 = 1.2$, $b = 0.75$) [8] and Anserini's defaults ($k_1 = 0.9$, $b = 0.4$) [11] on the remaining 400 Train queries (nDCG@5 of 0.473 vs. 0.454 and 0.436 on papers, and 0.371 vs. 0.354 and 0.344 on paragraphs). The tenfold difference in k_1 reflects document length: in a paper averaging 3,361 tokens, repeated occurrences of a term indicate its topic, whereas paragraphs, with a median of 40 tokens, rarely repeat a term, so whether a term occurs at all matters more.

2.3 Training Data

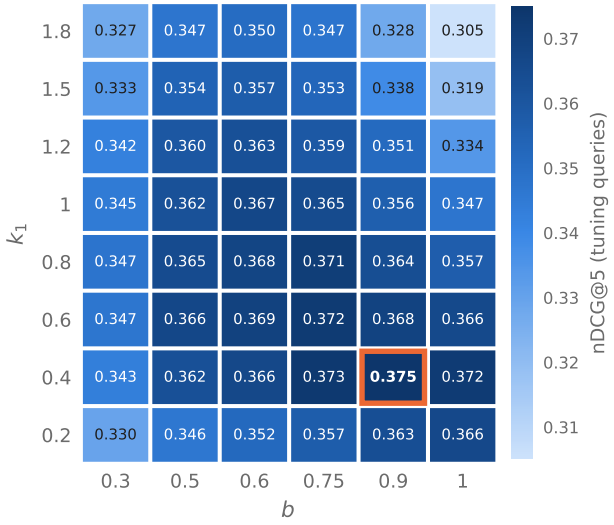
We fine-tune both neural models on Train queries only; Study and Test queries are never used for training or model selection. The 2,000 Train queries are split by query into 1,800 training and 200 validation queries, stratified by difficulty so that both sets keep the 30/50/20 mix of easy, medium, and hard queries, and no validation query appears in training. Every paragraph that was judged relevant or partially relevant serves as a positive, giving 5,138 positives for the training queries and 582 for the validation queries, about 2.9 per query.

The Train judgments contain no non-relevant (grade 0) paragraphs, so negatives must be mined. We use BM25 hard negatives: unjudged paragraphs that BM25 retrieves for the query. They share vocabulary and topics with it and are therefore more informative than random paragraphs. Because judgments are incomplete, an unjudged paragraph near the top of BM25's ranking is often relevant, so we skip BM25's top 20 results and sample 30 negatives per query at random from ranks 21–100, with a fixed seed per query. Random sampling gives each query a mix of close and easier negatives rather than only the hardest ones. Each training example is a (query, positive, negative) triple, one per positive, cycling through the query's negatives. Both neural models are trained with a contrastive loss in which each query's positive must outscore its own hard negative as well as the positives and negatives of the other queries in the batch (Sections 2.4 and 2.5).

Both models select their best checkpoint by nDCG@10 on the validation queries. Encoding all 2.28 million paragraphs at every evaluation would take 30 to 60 minutes (Section 3.7), so validation instead ranks a fixed sample of the collection: every validation positive, the top 100 BM25 results for each validation query, and 10,000 random paragraphs. Scores on this sample are used only to compare checkpoints, not to report effectiveness.



(a) Papers.



(b) Paragraphs.

Figure 1: BM25 nDCG@5 on 1,600 Train tuning queries. Outlined cells mark the selected settings.

2.4 Learned Sparse Retrieval

2.5 Dense Retrieval

2.6 Paper Ranking

2.7 Rank Fusion

BM25, SPLADE, and the dense model match queries to documents in different ways, through exact terms, weighted and expanded terms, and a single embedding, so their errors may differ as well. To test whether they complement each other, we combine their rankings with reciprocal rank fusion (RRF) [3], which scores each

document by its rank in each run:

$$\mathrm{RRF}(d) = \sum_{m \in M} \frac{1}{k + \mathrm{rank}_m(d)}, \qquad k = 60, \tag{3}$$

where M is the set of fused runs and $\mathrm{rank}_m(d)$ the rank of d in run m; a document absent from a run contributes nothing for that run. RRF uses only ranks, so it needs no normalization of the models' differently scaled scores, and the constant k limits how much a single top rank can dominate. We use the standard $k = 60$ without tuning. We fuse the top-100 runs of BM25 and SPLADE, and of all three models, separately for each collection, and keep the top 100 fused results per query. For papers, the BM25 paper run is fused with the neural models' MaxP paper runs. Fusion is evaluated on Study queries only and is not among our submitted Test runs.

2.8 Evaluation Setup

We evaluate each run's top 100 results with six metrics: nDCG, nDCG@5, MRR, P@5, P@10, and Recall@100, computed with ranx [1]. For a ranking with relevance grades $rel_i \in \{0, 1, 2\}$ at ranks $i = 1, 2, \ldots,$

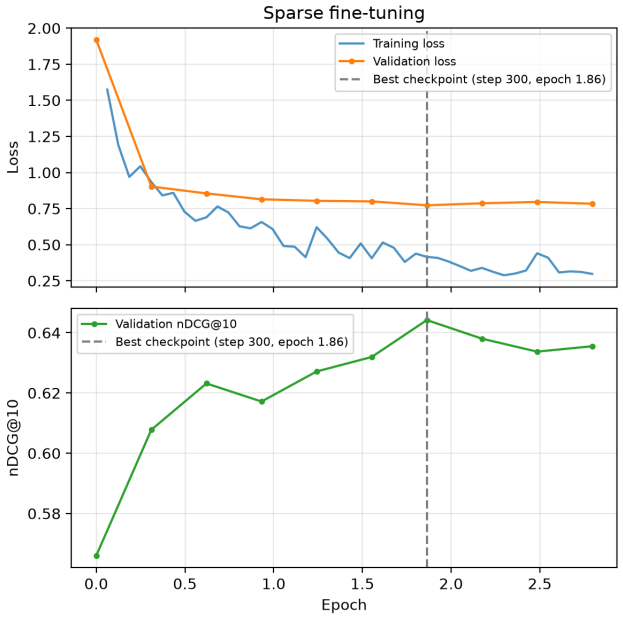
$$\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)}, \qquad \mathrm{nDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k}, \tag{4}$$

where IDCG@k is the DCG of an ideal ranking of the judged documents [6]; nDCG without a cutoff uses the full top 100. MRR is the mean over queries of $1/r$, where r is the rank of the first relevant document (0 if none is retrieved), P@k is the fraction of the top k documents that are relevant, and Recall@100 is the fraction of a query's relevant documents that appear in the top 100. Recall@100 is new in this phase: it measures how much a model could contribute to fusion or to a later reranking stage, beyond its top ranks. Documents absent from the judgments count as non-relevant.

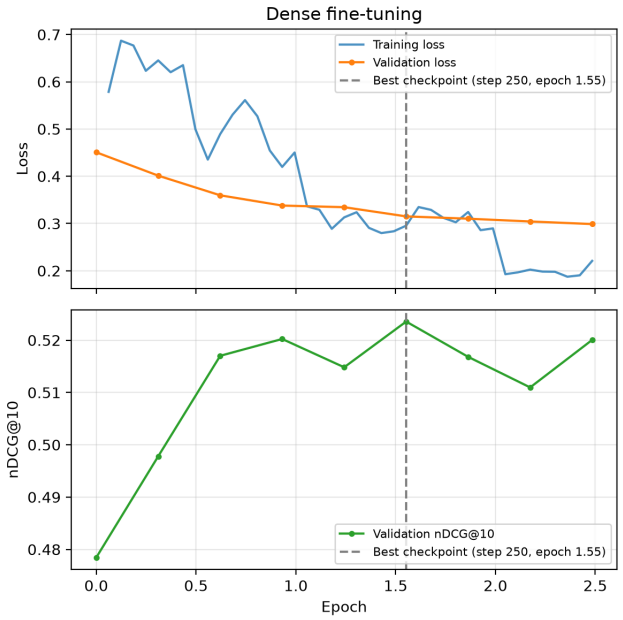
nDCG uses the relevance grades directly as gains. For MRR, P@k, and Recall@100, documents of grade 1 or 2 count as relevant. We also report strict MRR and P@5, in which only grade-2 documents count as relevant; these are averaged over the 238 Study queries that have at least one grade-2 document.

We test differences between runs with a two-sided paired Student's t-test over queries at $p < 0.05$, using ranx, between every pair of runs in a table: BM25, SPLADE, and the dense model; the zero-shot and fine-tuned version of each neural model; and each fused run against BM25, SPLADE, and the other fused run. We do not correct for multiple comparisons, so with many tests some individual significant differences may be by chance; we therefore emphasize differences that are consistent across metrics and collections.

The three query sets serve distinct purposes. Train queries are used for tuning BM25 (Section 2.2) and for training the neural models as well as selecting their checkpoints (Section 2.3). The Study queries are used for all reported results, and the Test queries, whose judgments are withheld, are for our submitted runs and manual annotation (Section 3.6).



(a) SPLADE.



(b) Dense (MiniLM).

Figure 2: Training and validation loss.

Table 1: Effectiveness on the 250 Study queries. Strict metrics count only grade-2 documents (238 queries). Superscripts mark a significant improvement over the model with that letter (paired t-test, $p < 0.05$).

	Model	nDCG	nDCG@5	MRR	P@5	P@10	R@100	Strict MRR	Strict P@5
Papers									
a	BM25	0.582	0.514	0.544	0.137	0.080	0.803	0.535	0.124
b	Dense	0.640^{a}	0.592^{a}	0.631^{a}	0.156^{a}	0.084	0.807	0.606^{a}	0.141^{a}
c	SPLADE	**0.671**^{a}	**0.631**^{ab}	**0.658**^{a}	**0.167**^{ab}	**0.090**^{ab}	**0.825**	**0.645**^{a}	**0.151**^{ab}
Paragraphs									
a	BM25	0.474	0.416	0.516	0.174	0.107	0.592	0.450	0.115
b	Dense	0.447	0.377	0.477	0.175	0.104	0.607	0.415	0.115
c	SPLADE	**0.494**^{b}	**0.434**^{b}	**0.539**^{b}	**0.190**	**0.116**^{b}	**0.630**^{a}	**0.484**^{b}	**0.135**^{ab}

Table 2: Zero-shot versus fine-tuned neural models on Study queries. † marks a significant difference between the two versions of a model (paired t-test, $p < 0.05$).

	Papers					Paragraphs				
Model	nDCG	nDCG@5	MRR	P@5	R@100	nDCG	nDCG@5	MRR	P@5	R@100
Dense, zero-shot	0.635	0.591	0.619	0.166†	0.812	0.458	0.385	0.484	0.174	0.633
Dense, fine-tuned	0.640	0.592	0.631	0.156	0.807	0.447	0.377	0.477	0.175	0.607
SPLADE, zero-shot	0.668	0.625	0.654	0.161	0.815	0.485	0.431	0.524	0.186	0.603
SPLADE, fine-tuned	0.671	0.631	0.658	0.167	0.825	0.494	0.434	0.539	0.190	0.630

3 Evaluation Results

3.1 Training

3.2 Effectiveness

3.3 Effect of Fine-Tuning

3.4 Rank Fusion

3.5 Query Difficulty

Table 4 breaks nDCG@5 down by query difficulty. On papers, the neural models improve most on easy and medium queries: SPLADE raises nDCG@5 over BM25 by 0.155 on easy and 0.133 on medium queries, but by only 0.023 on hard queries, where neither model differs significantly. It is only with fusion that we see a significant difference in nDCG (0.276 RRF vs. 0.187 BM25).

On paragraphs, BM25 remains the best single model on easy queries (0.676), significantly ahead of the dense model (0.559) and numerically ahead of SPLADE (0.627). Easy queries tend to reuse the wording of their source apragraph, which exact term matching rewards. SPLADE leads on medium queries (0.475 vs. 0.425 for BM25, not significant), and fusion is significantly better than every single model (0.547). On hard paragraph queries, all models score close to zero: 0.003 for BM25 and about 0.04 for the neural models. Both neural models are significantly better than BM25 on nDCG, but none reliably places a judged relevant paragraph in its top five. Incomplete judgments may contribute to this: in Part 1, BM25's top-ranked paragraph was unjudged for every hard Study query. Our Test annotations (Section 3.6) give a fuller picture of how the models perform on hard queries.

3.6 Test Annotation

3.7 Efficiency

4 Results Analysis

4.1 Comparing the Neural Models

4.2 Comparison with the BM25 Baseline

Comparing our Part 2 models with the Part 1 baseline on papers conflates two changes: the neural models use a different matching model, and they also rank papers differently, by their best paragraph rather than by the paper's full text. To separate the two, we also ranked papers with BM25 by MaxP, retrieving 2,000 paragraphs with the paragraph-tuned BM25 (Section 2.2) and keeping each paper's best paragraph (Table 7). This alone raises BM25's paper nDCG from 0.582 to 0.645. In fact, it significantly increases on every metric except P@10 and Recall@100. BM25-MaxP is statistically indistinguishable from the dense model on every metric. The dense model's advantage over BM25 on papers therefore comes from ranking papers by their best paragraph, not from dense matching, which is consistent with its weaker paragraph results. SPLADE remains ahead of BM25-MaxP (nDCG 0.671 vs. 0.645), but the difference is significant only for P@5; its advantage is concentrated on medium queries (nDCG@5 0.704 vs. 0.642, significant). With every model utilizing the same paper ranking method, SPLADE's improvement over BM25 becomes modest on both tasks, and the large gains in this phase come from MaxP and from fusion (Section 3.4).

MaxP helps most on easy queries, where BM25-MaxP reaches nDCG@5 of 0.830, the highest of any model. Many queries ask about a specific finding or method described in a single passage, so a paper's best-matching paragraph is a better indicator of relevance

Table 3: Reciprocal rank fusion on Study queries. Superscripts mark a significant improvement over the run with that letter (paired t-test, $p < 0.05$).

	Run	nDCG	nDCG@5	MRR	P@5	P@10	R@100	Strict MRR	Strict P@5
Papers									
a	BM25	0.582	0.514	0.544	0.137	0.080	0.803	0.535	0.124
b	SPLADE	0.671^{a}	0.631^{a}	0.658^{a}	0.167^{a}	0.090^{a}	0.825	0.645^{a}	0.151^{a}
c	RRF(BM25+SPLADE)	0.675^{a}	0.628^{a}	0.659^{a}	0.170^{a}	0.094^{a}	0.834^{a}	0.646^{a}	0.149^{a}
d	RRF(all three)	$\mathbf{0.701}^{abc}$	$\mathbf{0.655}^{ac}$	$\mathbf{0.689}^{ac}$	$\mathbf{0.174}^{a}$	$\mathbf{0.095}^{ab}$	$\mathbf{0.853}^{abc}$	$\mathbf{0.673}^{ac}$	$\mathbf{0.154}^{a}$
Paragraphs									
a	BM25	0.474	0.416	0.516	0.174	0.107	0.592	0.450	0.115
b	SPLADE	0.494	0.434	0.539	0.190	0.116	0.630^{a}	0.484	0.135^{a}
c	RRF(BM25+SPLADE)	0.531^{ab}	0.468^{ab}	0.569^{a}	0.201^{a}	0.119^{a}	0.666^{ab}	0.508^{a}	0.136^{a}
d	RRF(all three)	$\mathbf{0.550}^{abc}$	$\mathbf{0.489}^{abc}$	$\mathbf{0.582}^{ab}$	$\mathbf{0.218}^{abc}$	$\mathbf{0.130}^{abc}$	$\mathbf{0.689}^{abc}$	$\mathbf{0.523}^{ab}$	$\mathbf{0.148}^{abc}$

Table 4: nDCG@5 on Study queries by difficulty (75 easy, 125 medium, 50 hard). Superscripts mark a significant improvement over the model with that letter within the same group (paired t-test, $p < 0.05$).

		BM25 (a)	Dense (b)	SPLADE (c)	RRF (d)
Papers	Easy	0.635	0.771^{a}	0.790^{a}	$\mathbf{0.804}^{a}$
	Medium	0.571	0.639	0.704^{ab}	$\mathbf{0.717}^{ab}$
	Hard	0.187	0.207	0.210	$\mathbf{0.276}^{a}$
Paragraphs	Easy	0.676^{b}	0.559	0.627^{b}	$\mathbf{0.694}^{bc}$
	Medium	0.425	0.404	0.475^{b}	$\mathbf{0.547}^{abc}$
	Hard	0.003	0.036^{a}	**0.041**	0.036^{a}

Table 5: Grades of the top-ranked paragraph for the 250 Test queries, by difficulty: fine-tuned SPLADE (Part 2) and BM25 (Part 1). For 79 queries both models ranked the same paragraph first, and its Part 1 grade was reused.

Grade	Easy	Medium	Hard	All
SPLADE (Part 2)				
Relevant (2)	47	54	12	113
Partially relevant (1)	19	51	20	90
Non-relevant (0)	9	20	18	47
Relevant or partially	88%	84%	64%	81%
BM25 (Part 1)				
Relevant (2)	49	48	7	104
Partially relevant (1)	21	57	15	93
Non-relevant (0)	5	20	28	53
Relevant or partially	93%	84%	44%	79%

than its full text, in which a long paper can accumulate matches for common query terms across many unrelated paragraphs, as seen with query 2112 in Part 1. On hard queries the opposite holds: BM25-MaxP is the weakest model (0.141), and full-paper BM25 is significantly better on nDCG (0.256 vs. 0.201). When a query paraphrases its source passage, no single paragraph shares many terms with it, but the paper as a whole may.

Table 6: Index size, indexing time, startup time and average query latency on the 250 Study queries (papers / paragraphs). All times were measured on one shared server with an Intel Core i7-11700 CPU (2.50 GHz), 31 GiB of RAM and an NVIDIA RTX A5000 GPU (24 GB); encoders run on the GPU and index search on the CPU. Query latency includes query processing and, for neural paper runs, retrieving 2,000 paragraphs and MaxP. Neural models index only paragraphs. With a checkpoint every 50,000 paragraphs, dense indexing took 70.1 minutes.

Model	Index	Indexing	Startup (s)	Query (ms)
BM25	242 MB / 974 MB	81 s / 175 s	15.6 / 34.6	7.3 / 62.4
Dense	3.3 GB	31.8 min	64.5 / 41.0	231.6 / 230.1
SPLADE	2.8 GB	63.4 min	97.3 / 69.1	536.5 / 490.9

Table 7: Paper retrieval on Study queries when BM25 also ranks papers by their best paragraph (BM25-MaxP), as the neural models do. The last three columns give nDCG@5 by difficulty. Superscripts mark a significant improvement over the model with that letter (paired t-test, $p < 0.05$).

	Model	nDCG	nDCG@5	P@5	R@100	nDCG@5 Easy	nDCG@5 Med.	nDCG@5 Hard
a	BM25	0.582	0.514	0.137	0.803	0.635	0.571	0.187
b	BM25-MaxP	0.645^{a}	0.598^{a}	0.152^{a}	0.788	$\mathbf{0.830}^{a}$	0.642	0.141
c	Dense	0.640^{a}	0.592^{a}	0.156^{a}	0.807	0.771^{a}	0.639	0.207
d	SPLADE	$\mathbf{0.671}^{a}$	$\mathbf{0.631}^{ac}$	$\mathbf{0.167}^{abc}$	**0.825**	0.790^{a}	$\mathbf{0.704}^{abc}$	**0.210**

Query 2011, which BM25 failed in Part 1, shows where neural matching does help: *"Why is it harder to identify specific heart rhythm abnormalities using wrist-worn optical sensors than with electrical cardiac monitoring, and what complicates this process in wearable devices?"* Its relevant paper describes this in technical terms ("arrhythmias", "PPG", "ECG", "smartwatch"), and none of its five judged paragraphs appear in BM25's top 100. Both neural models retrieve them. The dense model ranks three partially relevant paragraphs, including the abstract, 3rd, 4th, and 6th, and SPLADE

ranks the fully relevant paragraph 8th and a partially relevant one 9th. On papers, the relevant paper rises from 12th with BM25 to 3rd with the dense model and 5th with SPLADE. The two models also succeed differently: the dense model places the fully relevant paragraph only 55th, while SPLADE places it 8th but ranks the abstract only 54th. Such disagreements are what fusion can exploit, since it rewards documents that several models rank well. Query 2011 is a hard query, so it also shows that the near-zero hard-query averages in Table 4 hide individual successes of this kind.

4.3 Successful Retrieval Cases

4.4 Failure Cases

4.5 Why Fine-Tuning Gave Little

4.6 Why Fusion Helps

4.7 Strengths and Weaknesses

5 Conclusion

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